

Francesco Vicidomini

Software Development Specialist with 6+ years of experience designing Angular and TypeScript applications for cybersecurity, European research, and public-sector platforms. Specialized in reactive architectures with RxJS and NgRx, with hands-on experience in technical decision-making, team coordination, mentoring, and international project collaboration.



CONTACT

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CORE SKILLS

Core Frontend

Angular TypeScript
RxJS NgRx Nx
Tailwind CSS

Web Platform & Quality

REST APIs WebSocket
React / Next.js (portfolio)
Jest Unit Testing

Backend

Node.js Spring Boot
MongoDB MySQL

DevOps & Practices

Git GitLab Docker
CI/CD AWS OAuth2
JWT Keycloak
Agile Scrum

Domains

Cybersecurity
Privacy Risk Assessment
Digital Twin
Public Administration

LANGUAGES

Italian Native
English B2

HOBBIES

Gym Books
Comics Video Games

PROFESSIONAL EXPERIENCE

Software Development Specialist

Jul 2023 – Present

Engineering Ingegneria Informatica · Naples, Italy

- Own front-end implementation decisions and technology selection for Angular cybersecurity platforms developed within EU-funded research projects.
- Collaborate in English with international consortia and present implemented features, technical progress, and proposed solutions during project meetings.
- Investigated redundant calculations in an existing system and guided an optimization that improved API response times without changing the expected results.
- Organize development work through GitLab Issues and helped simplify a heavily branched Git history into a clearer commit chain, making the evolution of implemented features easier to trace.
- Coordinate the development work of up to two colleagues and mentor them through solution analysis, trade-off evaluation, and context-aware technical decisions.
- Design maintainable reactive applications with Angular, TypeScript, RxJS, NgRx, Nx, REST APIs, and WebSocket integrations.

Software Engineer

Jun 2020 – Jul 2023

NTT DATA · Naples, Italy

- Developed an Angular platform supporting administrative processes for citizens, businesses, and tax assistance centres within an Italian province.
- Modernized legacy form workflows by introducing Angular reactive forms and FormControl-based validation.
- Created reusable modules and shared components that reduced duplicated implementation work and accelerated the delivery of similar user interfaces.
- Worked directly with the client to translate complex administrative requirements into clear, easy-to-complete interfaces and coordinated a three-person development team.
- Designed the front-end integration of a major new module, defining actors, permissions, restricted application areas, and reusable versus domain-specific components.

SELECTED PROJECTS

AVANT

2025 – Present

Engineering Ingegneria Informatica

NextGenerationEU cybersecurity project for Digital Twin systems. Designed and developed workflows to monitor a Digital Twin, model business cases and connected assets, simulate MITRE ATT&CK attack paths and countermeasures, and compare inherent and residual asset risks. Researched the Digital Twin domain to translate emerging concepts into an implementable platform solution.

Technologies: Angular 22, TypeScript, RxJS, NgRx, Nx, WebSocket, Spring Boot, MongoDB, Docker, Keycloak, CI/CD

SELECTED PROJECTS

ENCRYPT

2023 – 2025

Engineering Ingegneria Informatica

Horizon Europe privacy risk assessment project developed within a consortium of organizations from across Europe. Built a web module that models domain-specific asset taxonomies, evaluates LINDDUN privacy impacts and exposure to MITRE ATT&CK techniques, calculates privacy risk scores, and summarizes assessment results through interactive charts.

Technologies: Angular 22, TypeScript, RxJS, NgRx, Nx, Tailwind CSS, Node.js, MongoDB, Docker, Keycloak, CI/CD

PAB Goes Digital

2020 – 2023

NTT DATA

Large Angular platform for digitizing provincial administrative processes used by citizens, businesses, and tax assistance centres. Introduced reactive forms, reusable modules, role-based access patterns, and shared components while working directly with the client in an Agile Scrum team.

Technologies: Angular, TypeScript, Reactive Forms, Spring Boot, MySQL

FrancescoViciomini.dev

Personal project

Personal website used to experiment with modern web technologies and evolving front-end approaches. Originally developed with Angular, later rebuilt with React, and currently implemented with Next.js. Deployed on Vercel to gain hands-on experience with continuous integration and deployment workflows.

Technologies: Next.js, React, Angular, TypeScript, Vercel, CI/CD

[GitHub repository](#)

EDUCATION

Master of Science in Computer Science

2018 – 2020

Università degli Studi di Salerno · Salerno, Italy

Grade: 110/110 cum laude

Bachelor of Science in Computer Science

2013 – 2017

Università degli Studi di Salerno · Salerno, Italy

Grade: 94/110

ACADEMIC PROJECTS

Knoxly

Master's thesis privacy assistant for Telegram Web and Gmail. Built a two-stage text-classification pipeline using multilingual sentence embeddings and topic-specific sensitivity classifiers, exposed through a Flask API and integrated with a configurable Chrome extension.

Python, TensorFlow Hub, Multilingual Universal Sentence Encoder, Text Classification, Flask REST API, Chrome Extension

ShareErasmus

Erasmus platform developed while acting as project manager for a team of eight people.

Node.js, MySQL, Firebase, Jest, AWS

YASPL3

Designed a custom programming language and implemented its compiler in Java SE.

Java SE, Compilers, Language Design

N-body Simulation

Parallel and concurrent N-body simulation with AWS cluster configuration and performance testing.

C, OpenMP, AWS EC2

PUBLICATIONS

Ex Machina: Analytical Platforms, Law and the Challenges of Computational Legal Science

2018

Future Internet, MDPI · Academic research contribution

Authors: Nicola Lettieri, Antonio Altamura, Rosalba Giugno, Alfonso Guarino, Delfina Malandrino, Alfredo Pulvirenti, Francesco Viciomini, Rocco Zaccagnino

Implementation of a Tool for Risk Assessment of Personal Data Processing Using a Methodology Based on the LINDDUN and MITRE ATT&CK Frameworks

2025

ENCRYPT Project · Technical article · Engineering Ingegneria Informatica

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